Metamorphic testing of large language model question answering systems

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ABSTRACT

Large language models are capable of producing a wide variety of creative outputs, and their question-answering capabilities are compelling. Current testing efforts for large language models question answering systems are largely based on the reference paradigm and rely on manual labeling of test inputs. Testing of Large language models suffers from the test oracle problem. Metamorphic testing can test software by defining the expected relationships between inputs and outputs to ensure that the software still works correctly in different situations. In this paper, we propose a metamorphic testing method for large language models question answering systems. Specifically, starting from the premise of semantic similarity, metamorphic relationships are defined at three granularities: character, word, and sentence, and derived test cases are generated based on the metamorphic relationships. Meanwhile, the corresponding metamorphic relations based on prompt engineering are defined according to the characteristics of prompt engineering. The robustness of the large language models question answering system is tested by checking whether the prior metamorphic testing relationship is satisfied between the source and derived outputs of the large language models question answering system. To verify the effectiveness of the method, we conduct experimental validation on different large language models and calculate the metamorphic relation violation rate, which provides insights for studying the question answering capability of large language models.

Keywords: Metamorphic Testing, Large Language Model, Metamorphic relationship, Prompt engineering

1 INTRODUCTION

Software testing, as a key discipline in software engineering, is the core link to guarantee software quality and improve software reliability. In the field of artificial intelligence, Large Language Models(LLMs) with their excellent data processing efficiency and complex knowledge presentation capabilities have become a key tool for overcoming complex problems and driving technological innovation. In particular, the ability of LLMs to generate all kinds of highly creative outputs makes their question answering capabilities highly visible and a major highlight in the development process of current artificial intelligence. Currently, LLMs are already providing a wide range of generalized applications in various domains. Specifically they have a wide range of applications, not only limited to the directly relevant nature language processing (NLP) domains such as search engines [1], customer support[2] and translation[3], but also extending to a number of different scenarios such as code generation [4], healthcare [5], finance[6] and education[7]. This cross-industry and cross-context applicability highlights their flexibility and great potential for simplifying language-related tasks.

While the performance of Large language models is impressive, like other typical question answering systems, there are flaws in real-world application scenarios. As a real-world example, when asked a multiple-choice question with four options such as "The missing page scheduling algorithm is incorrectly described as ()" four times, the Large language models gave different answers in three of the four times. Without knowing the correct answer, such a wrong answer is very misleading to the user. In order to avoid these errors, large language models also require some very strict testing to ensure quality.

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Eighth International Conference on Artificial Intelligence and Pattern Recognition
(AIPR 2025), edited by Hui Tian, Proc. of SPIE Vol. 13993, 139935I
© 2025 SPIE · 0277-786X · doi: 10.1117/12.3093102

Current question answering system testing relies on manual labeling, which is costly, time-consuming, and may not immediately address bugs. And there has not been much research on robustness testing for LLMs question answering systems. To address these issues, Chen et al.[8] proposed the technique of QAAskeR for question answering systems, which is free from the reliance on manual annotation. QAAskeR constructs pseudo-facts by combining the input questions with the resulting answers, a process that lurks in a number of challenges: firstly, the generated questions may not be valid, and secondly, the new questions may not be present in the knowledgebase or the associated context, making the questions difficult to answer. In addition, generative large-scale models often lack reference links to articles and option settings, and they directly receive user questions and return answers, making it difficult to pre-evaluate the difference between the expected output of an answer and the actual result.

To address these issues, this paper tests LLMs question answering systems using Metamorphic testing, which aims to identify potential vulnerabilities in the question answering capabilities of these LLMs in order to improve the reliability of LLMs.

The main work accomplished in this research:

I Exploring a new testing technique for LLMs: Firstly, the aim of this research is to work out an effective methodology for evaluating LLMs that reflects the problems that users may encounter in the real world.

II Designing metamorphic testing experiments: Designing to utilize Metamorphic testing to assess the question answering ability of LLMs while improving the validity of the test.

III Share the performance of different LLMs: Conduct experiments on different versions of LLMs to understand their question answering capabilities.

2 BACKGROUND

2.1 Large language models

The development of Large language models began in the 1950s. It has gone through three stages of development: simple statistical language models[9] (SLMs), neural language models[10] (NLMS), and pre-trained language models[11][12] (PLMs). The Large language models stage is generally based on the Transformer architecture, which further improves the generation and comprehension capabilities through larger scale pre-trained data and model parameters. Large language models are now able to analyze the context based on the context, understand the real intention of the question, and make a reasonable answer. For questions that require reasoning, Large language models are also able to give more reasonable answers through analysis. As time continues to pass, they are able to automatically adjust their internal knowledge structure and expression.

2.2 Prompt engineering

Prompt engineering emerged as a way for LLMs to be able to generate specific textual outputs with specific expectations based on inputs. Through careful design and improvement, the model can be guided to generate accurate and useful output that better fits the scenario. The core of this approach lies in influencing the model's output behavior by changing the input prompts, thus achieving flexible regulation of the model's behavior without changing the structure or parameters of the model itself[13].

2.3 Metamorphic testing

In 1998, Chen et al.[14] proposed metamorphic testing, which uses metamorphic relations (MRs) to generate new test cases and verify that the results of their execution conform to the relations to determine program correctness. Metamorphic relations define association rules under input-output transformations, such as the periodicity of sin (x) $\sin(x) = \sin(x + 2\pi)$. Metamorphic testing is designed to find problems that were missed by the original testing, and outputs that do not match the metamorphic relationship may indicate a bug in the program.

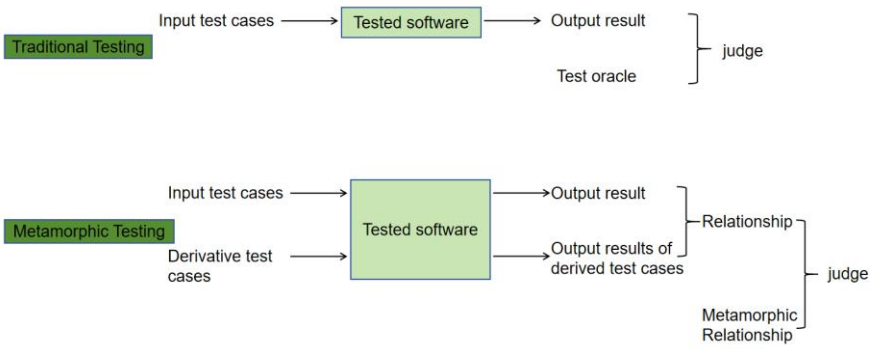


Figure 1. Differences between traditional and Metamorphic testing

3 METAMORPHIC TESTING FRAMEWORK FOR LLMS

3.1 The metamorphic testing framework

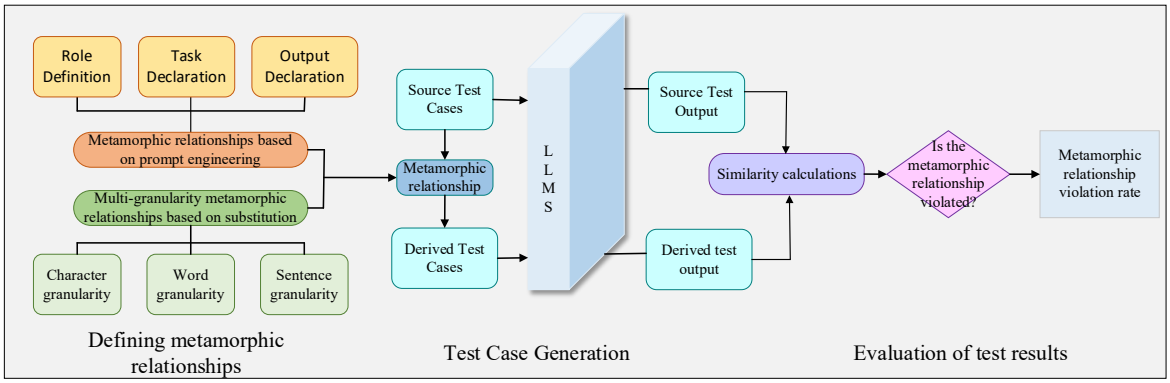


Figure 2. Differences between traditional and Metamorphic testing

While the previous chapter described the foundation, this chapter delves into the core methodology. Based on the principle of metamorphic testing, we design a framework for testing LLMs question answering system (see Figure 2). There are three phases: 1) Define metamorphic testing relationship, explore the impact of prompt and input changes on system robustness, and design two relationships; 2) Test case generation, generate derivative cases based on source input and metamorphic relationship, and obtain answers; 3) Answer evaluation, compare the semantic similarity of answers, calculate the violation rate of metamorphic testing, and evaluate the robustness. This framework effectively evaluates the robustness of the system by defining relations, generating use cases, and evaluating answers, does not rely on labeling information, supports multi-party evaluation, and provides a new way of thinking for the quality evaluation of intelligent question answering systems. At its simplest, when asking them the same question containing multiple options multiple times, we want to get answers that should be consistent, just as police officers record statements from suspects during interrogation, and if the transcripts are inconsistent, it may be time to consider whether the suspects have lied. It is important to perform such tests on LLMs given their real-world usage.

3.2 Defining the metamorphic relationships

In order to explore the stability of Large Language Model when encountering semantically similar questions and the effect of prompt engineering on the answers generated by Large Language Model, this study mainly starts from the two defined metamorphic testing relationships to generate corresponding test cases to achieve the purpose of testing.

MR1:Does changing the prompts affect the ability of the LLMs to handle questions without changing the description or the question?

The prompt is divided into three parts: Role Definition (defining model roles, such as reading assistants), Task Declaration (declaring specific tasks), and Output Declaration (specifying output results). Role Definition and Task

Declaration enable the model to specify its functions and goals, while Output Declaration specifies the results that the model should produce. In the experiments, by adjusting the Role Definitions and Task Declarations, the effects of small changes on the model's response can be explored.

MR1-0: No changes were made to the description and questions while keeping the Role Definition and Task Declaration portions of the prompt. Since the description, questions, and prompts remain the same, the expected output should be identical to the original output. This metamorphic testing relationship is mainly used to validate the stability and consistency of the testing environment and to ensure that the test results are consistent without any changes.

MR1-1: Remove the Role Definition portion of the prompt and keep the Task Declaration portion without changing the description or questions. Although the Task Declaration remains the same, the lack of the Role Definition may create some minor differences in understanding during execution that probably should not affect the final output.

MR1-2: Without changing the description and questions, the Task Declaration portion of the prompt is removed and the Role Definition portion is retained. Since the Task Declaration is removed but the Role Definition is retained, the output of the LLMs should be sufficient to understand the Role Definition and produce the same output.

MR1-3: Remove both the Role Definition and Task Declaration portions of the prompts without changing the description and questions. While this may lead to a more complex and time-consuming testing process, LLMs should respond accordingly without any prompts.

MR2:Does changing the partial composition of a question without changing the prompt affect the ability of the LLMs to handle the question?

In the prompt-based metamorphic relationship, attention is given to model generalization, contextual understanding, and problem-solving flexibility. If the model cannot effectively adapt to problem variations, the generalization ability is questionable. LLMs should be flexible in handling problem variants to get the right answer, otherwise it is difficult to capture the core intent of the problem.

In generating test cases for this metamorphic relationship, we consider changes in granularity (char, word, sentence), operation (substitution, insertion), and logic (semantic invariance). The core idea is that semantically consistent questions should yield the same answer as the source question.

The metamorphic relation we define not only logically reasoned, but also deeply considered the complexity and diversity of the real world. We recognize that in real-world applications, LLMs often face a variety of challenges that may stem from noisy input data, differences in cultural backgrounds, and the diversity of user intentions. Therefore, we design metamorphic relationships to capture these real-world challenges and evaluate the stability and accuracy of the model in addressing them.

Specifically, these rules defined in this experiment reflect plausible variations of the inputs under different realistic scenarios, and it is expected that the model should keep the output consistent or conform to the expected regularity under these variations. Therefore, these metamorphic relationships are selected to simulate the user's usage in the real world.

Character Granularity:

MR2-1: Replace some characters of the source problem with characters that are visually more similar to it, with the expectation that the output should have similar semantics to the original output. These characters can be the alphabet of another language, or some other similar symbols. The expected output should be similar to the original output.

MR2-2: By replacing some characters of the source problem with some of its “splits”, the expected output should be semantically similar to the original output. The expected output should be similar to the original output.

MR2-3: Replace some characters of the source problem with some of its “combinations”, the expected output should be similar to the original output.

MR2-4: Replace some characters of the source question with irrelevant symbols such as “*”, the expected output should be similar to the original output. (Table 1 shows some examples.)

Word Granularity:

MR2-5: Replacing words in the source problem with other languages with the same semantics, the expected output may still be relevant or similar to the original output.

MR2-6: Replace words in the source problem with their synonyms. The expected output should be related or similar to the original output.

MR2-7: Replace the word in the source question with its scrambled order. The expected output should be related or similar to the original output.

MR2-8: Inserting spaces or other symbols into words in the source problem. The expected output should be related or similar to the original output.

Sentence Granularity:

MR2-9: Extend the source problem to a combination of the source problem and redundant sentences. Where the inserted redundant sentence should be a description that does not affect the answer to the original question, the expected output should be the same as the output obtained when using only the original question as input.

Table 1. Example of character granularity metamorphosis relationship

MRs	Example Source Questions	New Questions
MR2-1	what other books did ch**a**rles dickens write?	what other books did ch**ä**rles dickens write?
MR2-2	what kind gove**rn**ment does egypt have?	what kind gove**m**ment does egypt have?
MR2-3	what does ja**m**aican people speak?	what does ja**rn**aican people speak?
MR2-4	what are the songs that justin bieber wrote?	what are the songs that j*stin b*eber wrote?

3.3 Test case generation

Character granularity: When selecting target words, the NLTK (Natural Language Toolkit)[15] package is used to perform operations such as deactivating words and then selecting the target words that need to be performed in the next step. For MR2-1 and MR2-2, DeepAI visual similarity API is used to find candidate replacement characters. MR2-1 finds the most visually similar characters in the Greek and German alphabets, and MR2-2 needs to find the most similar "split" method to replace the characters in the target word. Moreover, a dictionary is generated based on the original and candidate characters of MR2-2. If any combination of letters in the target word in MR2-3 appears in MR2-2's dictionary, then they are replaced. For the target word, MR2-4 does this by randomly replacing one of the vowel characters with "*".

Word granularity: The line of sight for metamorphic relations on word granularity i.e., adding a word granularity perturbation factor to the source problem generates a derived set of test cases. The word granularity perturbation factor that is added is by substituting minor perturbations such as other languages, synonyms, jumbles, etc., with the aim of enabling the Large Language Model to give consistent results for problems with minor differences when dealing with these problems. If random substitutions are made to the words in the problem, it may complicate the problem, cause it to be semantically incoherent, or even change the meaning of the original sentence, which will make the problem meaningless. When choosing the target words, some constraints need to be made, and related studies have shown that replacing some nouns in a sentence has less effect on the sentence.

When selecting a noun to be replaced, the DeepSelect[16] algorithm is used to add the noun constraint variant NounSelect, and the problem is put through BerkeleyParser[17] component analysis to obtain a syntactic component tree. Set the set of candidate paths, traversing each leaf node can be viewed as selecting a path that reaches that node, sorting the paths that reach each node, and selecting the shortest path length to record into the candidate set. Randomly select one of the paths whose endpoint is a noun as the location to be replaced.

MR2-5 replaces words to other languages using Google Translate API, which randomly replaces them to another language.MR2-6 synonym replacement uses Synonyms, a synonym replacement package, to find synonym substitutions.MR2-7 uses character swapping to replace the target word to its disordered case. For MR2-8, randomly inserting spaces or "*" in the target word has been used to achieve the replacement.

Sentence granularity: To perform a variation of sentence granularity, an original problem is selected, and then another redundant sentence is inserted into this sentence, combining together to form a new, longer problem. The inserted sentence is a fact in itself, but has no relation to the source question, so this derived question will still be semantically consistent with the source question. In order to make this derived question more coherent, it is common to join sentences using phrases such as "I hear that..." phrases such as "I hear that..." are often used to make the sentence more readable. Although the new question is more complex than the source question, it does not change the core semantics of the source

question. Therefore, the LLMs is dealing with such a question is that he should give the same answer to such two semantically consistent questions. The robustness of the LLMs when dealing with different types of problems is tested through the metamorphic testing of the sentence granularity relationship.

ALGORITHM 1: NounSelect

Input: source question *Question*, candidate set *Candidatenum*

Output: *Finalpath* to be replaced

Create a candidate set *Candidates* for the path to be replaced, initialize to the empty set.

Tree=BerkeleyParser(*Question*)

Traverse the leaf nodes of *Tree*, add each path of the leaf nodes to the path set PathSet, and sort the paths in descending order

For each *path* **in** *PathSet* **do**

if length(*Candidates*)<Candidatenum then

Candidates.append(*path*)

for each *path* **in** *Candidates*

Nonpath.append(path)

Finalpath=random.choose(*Candidates*)

return *Finalpath*

3.4 Evaluation of test result

Validating the results of LLMs metamorphic testing in this paper uses cosine similarity, which evaluates the degree of similarity between two vectors by calculating the cosine value of the angle between them. The magnitude of the cosine value is between [0,1] and the closer to 1, the more similar the two are. Assuming that there exist two vectors, where $A = (a_1, a_2, \dots, a_n)$,$B = (b_1, b_2, \dots, b_n)$, then and the cosine similarity of the two is calculated as formula (1).

$$Similarity_{A,B} = \frac{A \cdot B}{\| A \| \| B \|} = \frac{\sum_{i=1}^{n} A_i \times B_i}{\sqrt{\sum_{i=1}^{n} (A_i)^2} \times \sqrt{\sum_{i=1}^{n} (B_i)^2}} \quad (1)$$

In order to evaluate the effectiveness of metamorphic testing, the violation rate (VR) is used here to calculate the test results for evaluation purposes. Violation rate of metamorphic relationship refers to the proportion of new outputs and original outputs that violate the expectations defined by the metamorphic relationship when the inputs are changed according to the metamorphic relationship in metamorphic testing. Simply put, it is the percentage of test cases in the test set that violate the metamorphic testing relationship. The formula is as formula (2).

$$VR = \frac{V}{T_C} \times 100\% \quad (2)$$

T_C is the total number of test cases and V is the number of use cases violating the metamorphosis relation. VR indicates the extent to which the system violates the metamorphosis relation. A high value of VR indicates a poor performance of the system and a low value of VR indicates a high degree of conformity. $VR = 0$ implies that the system is robust under this metamorphosis relation but does not exclude other defects.

4 EXPERIMENTAL EVALUATION

4.1 Experimental setup

(1) Datasets

For the purpose of this study, the widely recognized WebQuestionsSP[18] and ComplexWebQ[19] are used here. Questions from these datasets will be used as the original test case input for this method in subsequent studies. Table2 shows the specifics of the datasets.

WebQuestionsSP is a classic dataset for evaluating open-domain question answering, Semantic Parsing, and Entity Linking, based on Freebase, with questions originating from real user queries and answers from Wikipedia, emphasizing the ability to deal with real-world complex problems. ComplexWebQ is an open source dataset for complex question

answering that is a challenging version of WebQuestionsSP, containing more multi-hop relationships and complex questions with constraints.

Table 2. Specifics of the datasets

Datasets	Training	Test	Total
WebQuestionsSP	3098	1639	4347
ComplexWebQ	31214	3475	34689

(2) Subjects

In our research work, two versions of GPT were chosen to represent LLMs to ensure a more nuanced understanding of their question answering capabilities.

GPT-3.5 Turbo: based on the Transformer architecture, it is a large-scale pre-trained language model. It is optimized in terms of processing speed and quality of generated text compared to GPT-3. The Turbo version aims to provide faster response times with lower computational resources while maintaining a high level of text generation capability.

GPT-4.0-Turbo: The latest generation of language models developed by OpenAI, further optimized based on the architecture of its predecessor, GPT-4. The model is designed to be more responsive and efficient, while maintaining excellent text generation quality. Experimental results

4.2 Experimental results and analysis

4.2.1 Impact of prompt engineering on Large language models

While keeping the question description and the question itself unchanged, it was explored how modifications to the prompt section would specifically affect the output or performance of the LLMs question answering system. To address this question, each question used to test the dataset was generated with derived inputs based on defined metamorphic relationships. By comparing the results of source and derived inputs answered by the LLMs, the results of the experiment are shown in Table 3.

Table 3. Impact of prompt changes on LLMs

	MR1-0（%）		MR1-1（%）		MR1-2（%）		MR1-3（%）	
Dataset	WQSP	CWQ	WQSP	CWQ	WQSP	CWQ	WQSP	CWQ
GPT-3.5 Turbo	2.93	3.32	2.93	3.55	2.92	3.43	3.51	3.64
GPT-4.0-Turbo	2.07	3.01	2.22	3.13	2.82	2.56	3.24	3.52

Based on the results of the experiment, the LLMs showed the best ability to process the questions most consistently when both Role Definition and Task Declaration were available. When either of the Role Definition and Task Declaration is missing, the results of the LLMs' answers may be somewhat different from the first output, which may be due to the fact that the LLMs need to be explicit about the task they are going to do and are more in need of having the ability to get a better answer for their role. The LLMs had the highest rate of metamorphic relationship violation when both Role Definition and Task Declaration were missing.

For different versions of GPT, GPT-4.0-Turbo outperforms GPT-3.5-Turbo, probably due to the fact that newer versions of the model usually have improved language comprehension, generative power, and logical reasoning, which has larger parameters, larger training data, and corresponding optimized algorithms that can perform better when dealing with the problem.

Although preliminary metamorphic testing of LLMs by defining prompts has investigated some of the performances, the research in this area is still relatively simple and needs to be progressed, such as proposing more complex prompt engineering to explore its performances and provide new perspectives for evaluating the validity and reliability of various models in real-world scenarios.

4.2.2 Impact of granularity changes on large language models robustness

In order to investigate the effectiveness of MR2, we conducts a large-scale empirical study where question answering consistency is evaluated on the dataset and the LLMs rows under test used in our experimental design. Corresponding to each question in the dataset, this paper generates a large number of derived questions at character granularity, word granularity, and sentence granularity, respectively. The corresponding question-answer results are obtained after feeding

them into the LLMs respectively, and the metamorphic relation violation rate is used as an evaluation metric. If a metamorphic relation is satisfied, it is scored as 1, otherwise it is scored as 0. The fact that the test sample violates a certain metamorphic relation indicates that the LLMs under test has poor question answering robustness to that metamorphic relation. A lower evaluation score indicates that the LLMs can effectively recognize these metamorphic relationships and give consistent answers for this metamorphic relationship. The results of the experiment are shown in Table 4.

Table 4. Effect of each granularity perturbation on LLMs question answering ability

	MR2-1 (%)		MR2-2 (%)		MR2-3 (%)		MR2-4 (%)	
Dataset	WQSP	CWQ	WQSP	CWQ	WQSP	CWQ	WQSP	CWQ
GPT-3.5 Turbo	7.75	7.21	6.61	3.28	6.83	7.29	6.93	9.54
GPT-4.0-Turbo	5.31	4.92	4.43	3.19	5.55	3.17	3.93	7.56

MR2-5 (%)		MR2-6 (%)		MR2-7 (%)		MR2-8 (%)		MR2-9 (%)	
WQSP	CWQ	WQSP	CWQ	WQSP	CWQ	WQSP	CWQ	WQSP	CWQ
11.25	19.63	31.12	35.09	43.83	44.23	9.10	10.33	9.21	10.68
8.28	13.10	27.88	30.18	42.96	40.41	8.00	3.56	6.23	7.33

From the experimental results, it can be seen that the Large Language Models have a low violation rate of the defined metamorphic relationship between character granularity and sentence granularity, which may be due to the fact that these Large Language Models are trained with a large amount of text data, which includes correctly spelled text. Through this large-scale training, the models learn the correct spelling as well as common language structures, and the Large Language Models have some contextual understanding to reason about the meaning of the wrong words based on the semantics and correct them when producing answers. However, when dealing with some flexible questions, the performance of the LLMs is slightly unstable, and the answers produced may vary, which affects the robustness of question answering.

5 SUMMARY

In this study, a new testing technique for large language models question answering system is explored, which can effectively alleviate the difficulties currently encountered in large models testing: the need for reference labeling, and the possibility that the generated questions may not be answerable. Finally, the validity of the methods is verified in different versions of large language models experiments and their capabilities are evaluated.

Although many metamorphic relationships have been identified for a variety of applications, most of them have been identified in an ad hoc and arbitrary manner, and the study of the links between different metamorphic relationships and the construction of a genealogy for them will be an important research direction in the future.

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